Telugu Government Form Intelligence Platform

A policy-controlled AI assistance layer for Telugu voice, Telugu-English documents, and faster operator-led form completion

Architecture baseline + local proof of concept

Assist, do not replace

Existing government applications remain the official systems of record.

Operator remains in control

AI prepares evidence-backed suggestions; review and final submit stay in the official workflow.

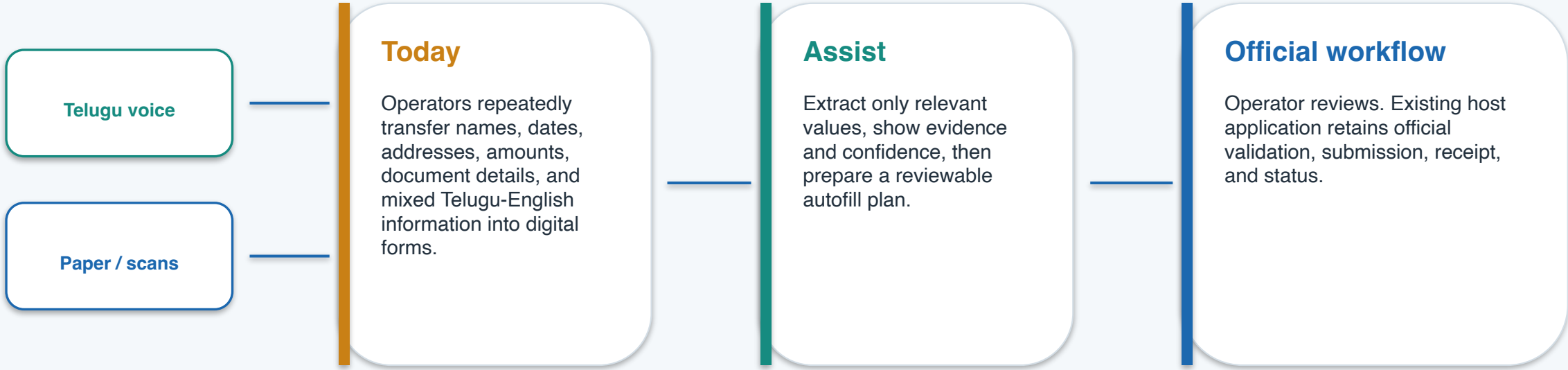
POC scope

Chrome sidecar for the Telangana MeeSeva registration page, running locally on one Mac.

For discussion with government authorities and technical stakeholders

The operational problem

The focus is faster, safer preparation of form values - not replacement of the official workflow.



No autonomous statutory decision.

No portal replacement.

No final submission without the official workflow.

Clear boundary: assist, do not replace

The platform has a deliberately constrained role in government-service workflows.

Platform-owned intelligence

- Telugu/Telugu-English speech-to-text
- Telugu-English document analysis
- Canonical field extraction
- Form mapping, relevance, confidence
- Approved document/service guidance
- One-click operator-initiated autofill
- Non-PII metrics and governed learning

Host-application ownership

- Operator identity and permissions
- Official form UI and draft
- Application persistence
- Official validation and statutory decisions
- Submission, receipt, status, certificates
- Citizen notice, retention, and access rights
- Rendered accessibility/GIGW evidence

The platform never owns credentials, payments, OTP/CAPTCHA, or official applications

Platform vision versus first proof of concept

The local POC proves the core loop. It is not a production or statewide deployment claim.

Target platform

Reusable self-hosted services

- Telugu ASR + Telugu-English OCR
- Form/document intelligence
- Policy-controlled integrations
- Versioned document/service packs
- PostgreSQL control plane
- Knowledge graph and controlled learning
- Optional API delivery adapters

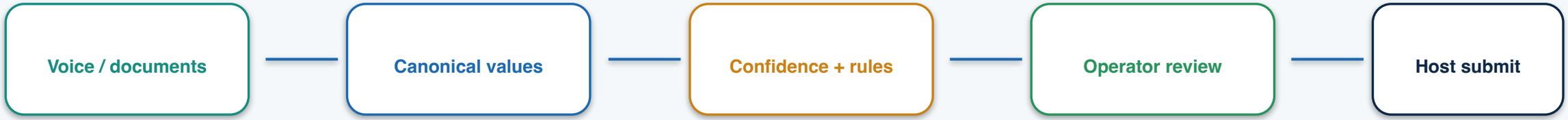
Initial local POC

Chrome sidecar on the Telangana MeeSeva user-registration page

- Runs on one Mac
- QR/local document intake
- Telugu voice for gaps
- One-click eligible-field fill
- Operator reviews and manually submits
- Simulated adapters replaced progressively after model evaluation

Operator control is the safety feature

Every suggestion is source-aware and reviewable. The platform never invents a missing value.



AUTO_SUGGEST

Eligible value, evidence available, and policy allows suggestion. Host retains correction capability.

REVIEW_REQUIRED

Medium confidence, conflicting source, learned alias, or high-risk field.

MANUAL / NO VALUE

No safe mapping or no relevant voice/ document value. Do not fill.

Passwords, secret answers, OTP/CAPTCHA, payment, signatures, declarations, consent, hidden fields, and submit controls are always excluded.

Two planes: ephemeral transaction data and governed control plane

Normal citizen transaction content is not the platform's database.

Ephemeral transaction data plane

Processes only for the active request/session

- Voice/audio
- Documents/photos/PDFs
- OCR/ASR text
- Extracted field values
- Form suggestions

Clears normal transaction content after response. No raw content in logs, metrics, traces, or learning stores.

PostgreSQL control plane

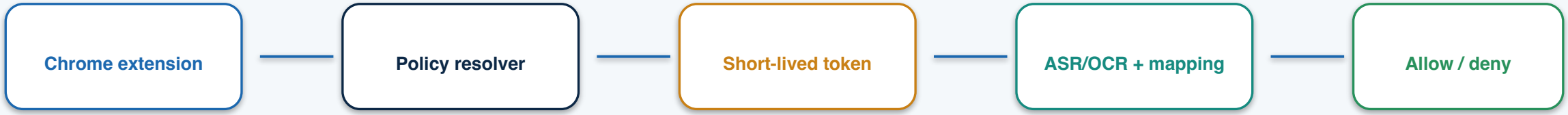
Durable non-transaction platform intelligence

- Integrations and allowlisted forms
- Form profiles and policy releases
- Model versions and pack versions
- Aggregate usage metrics
- Non-PII knowledge graph
- Reviewed learning candidates and governance events

No citizen profiles, applications, transcripts, OCR text, document bytes, or form values in control-plane tables

Security and policy enforcement

Assistance activates only on an approved integration and matching form profile.



POC policy scope

Tenant + local extension integration + environment + exact MeeSeva registration URL + static non-sensitive form fingerprint.

Default deny

If URL/fingerprint does not match, sidecar does not activate or fill fields.

Future expansion

Service/page/role policies first. Centre/mandal/area targeting is intentionally deferred.

Integration paths follow official API capability

The intelligence pipeline remains the same; the final delivery adapter changes.

No usable API

Chrome sidecar maps and fills eligible fields. Operator submits in the existing workflow.

Read-only API

Use official schema, allowed values, requirements, and reference data. UI remains the final submit path.

Draft/write API

Create/update an official host-owned draft after review. Official workflow keeps final submit.

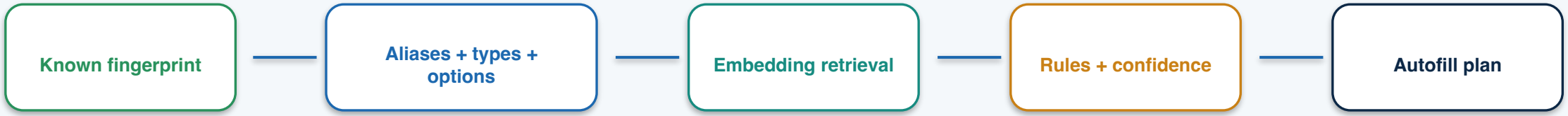
Authorized submit API

After explicit operator confirmation, call official API with idempotency, validation, and safe retry controls.

Direct API use never stores operator passwords. It uses an official delegated token, mTLS identity with verified operator context, or another approved government authorization mechanism.

Form intelligence: deterministic first, AI within guardrails

Normal autofill should not require an LLM call per field.



Normal path

Known labels/form profiles use exact rules, type constraints, section context, option matching, and cached mappings.

Unfamiliar labels

A small multilingual embedding model retrieves a limited candidate set for deterministic scoring.

Rare fallback

Small local LLM only for unknown-form review; outputs cannot directly fill, approve, or change configuration.

If no relevant and sufficiently confident source value exists, the result is NO_RELEVANT_VALUE - never a guess.

Models are evaluated before selection or rollout

Model names below are candidates, not production selections or quality claims.

Telugu / Telugu-English ASR

Candidate baselines

- Whisper large-v3 + faster-whisper
- AI4Bharat IndicConformer
- IndicWav2Vec as adaptation comparator

Evaluate Telangana dialect, code-switching, critical entity accuracy, correction effort, latency, and license provenance.

Telugu-English OCR

Candidate baselines

- PaddleOCR Telugu + PP-StructureV3
- Tesseract tel+eng
- PaddleOCR-VL for complex layouts

Evaluate document type, field exact match, numeric accuracy, evidence quality, latency, and resource use.

Release gate

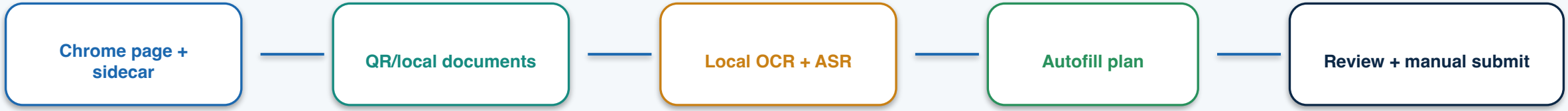
Required before pilot use

- Pinned source/runtime/license
- De-identified held-out evaluation
- Approved thresholds
- Failure/review route
- Rollback version
- No raw PII in telemetry

The local Mac CPU POC is for functional comparison; deployment latency must be validated on target hardware

The local POC: Telangana MeeSeva registration page

Purpose: prove a real multi-source form-assist loop without changing the host application or submitting an account.



May fill

Eligible visible profile/data fields: name, DOB, gender, address, city/state/pincode, mobile/email, and other reviewed data fields.

Must not fill

Passwords, secret answers, OTP/CAPTCHA, payment, signatures, declarations, consent, hidden fields, register/submit controls.

Local deployment

Extension, backend, models, QR upload session, temporary processing, and control-plane database run on one Mac.

QR upload and local scanner upload are analyzed first. Browser upload-field attachment is deliberately deferred from the initial POC.

Performance is a product requirement

The platform must beat keyboard-heavy entry, not add waiting time.

Interaction principle

Collect asynchronously
Show useful results progressively
Fill all eligible fields together

Never wait per field.
Never put a generative LLM in the normal autofill path.

Initial UX targets

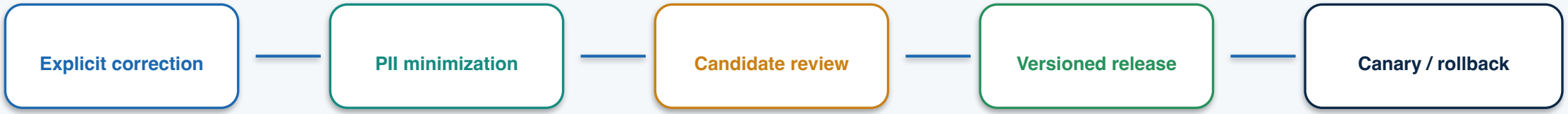
Known form recognition: < 200 ms
Sidecar/cached plan: < 300 ms
Map ready values: < 100 ms
Bulk fill: < 500 ms
Short voice final: target < 2 sec
First useful OCR: target < 3 sec on target hardware

How we achieve it

Warm models
Parallel document workers
Focused OCR regions first
Full-page fallback only when needed
Cached form rules and aliases
Progressive sidecar updates
Bulk field writes

Knowledge graph, learning, and metrics

The platform improves safely without becoming a shadow citizen-data store.



May retain after minimization

Approved Telugu-English aliases, document layout deltas, form-label mappings, confidence calibration, version/provenance, and aggregate quality signals.

Never retained as platform learning data

Raw documents/audio, names, Aadhaar, phones, application IDs, raw form values, free-text operator notes, or unapproved statutory changes.

POC mode: collect and review non-PII candidates only. No automatic promotion to runtime behavior.

GIGW, privacy, and operational boundary

The platform supports GIGW-aligned host applications; it does not claim GIGW or STQC certification.

Platform commitments

- Accessible sidecar/result contract
- Telugu/English semantic outputs
- Bounded uploads and retry-safe APIs
- Ephemeral transaction processing
- Integration policy/security
- Evidence and confidence explanations

Host responsibilities

- Official rendered form accessibility
- Language selection and content
- Citizen privacy notice/retention
- Operator authentication
- Secure hosting and audit
- Official submit/receipt/status

Before any production rollout

- Formal data-flow review
- Model/license approval
- Security and privacy controls
- Accessibility/mobile evidence
- Approved service/document rules
- Deployment/operations approval

Decisions requested and proposed next steps

A constrained, evidence-based evaluation path before any production commitment.

Authority / program ask

- Endorse operator-assist, non-replacement boundary
- Confirm use of registration-page POC for evaluation
- Nominate owner for future host integration discussion
- Confirm appropriate test-data and evaluation governance

Technical-team ask

- Review Chrome extension constraints
- Validate local/on-prem POC environment
- Advise on API/schema availability
- Identify official data/validation sources
- Review security and temporary-upload controls

Proposed sequence

1. Build local POC shell
2. Evaluate ASR/OCR candidates
3. Measure quality and latency
4. Demonstrate constrained autofill
5. Review evidence and risks
6. Decide whether to expand integration

Outcome sought: permission to evaluate safely, not a request to deploy or replace the existing system